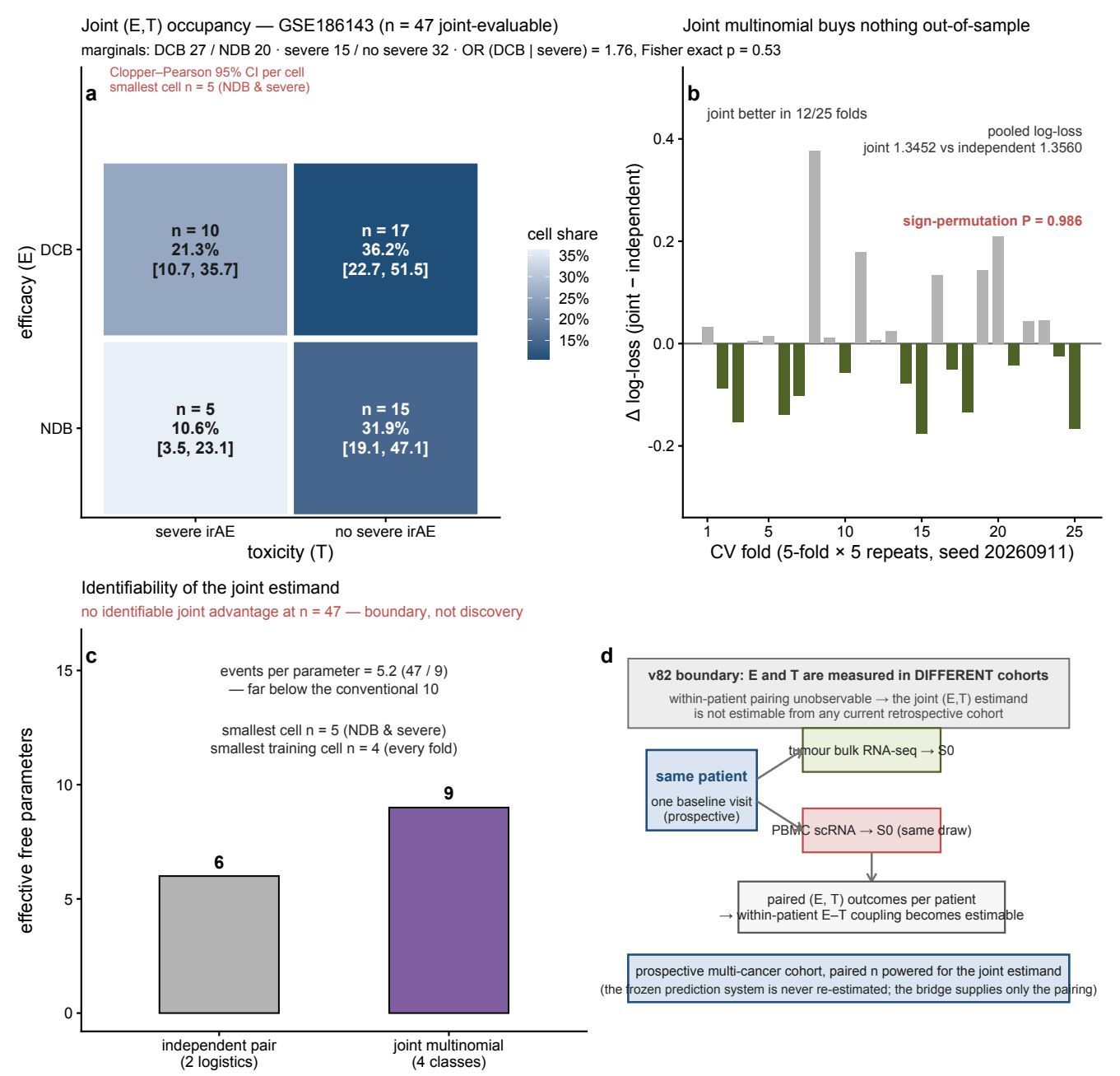

Joint (E,T) occupancy — GSE186143 (n = 47 joint-evaluable)
marginals: DCB 27 / NDB 20 · severe 15 / no severe 32 · OR (DCB | severe) = 1.76, Fisher exact p = 0.53
Clopper–Pearson 95% CI per cell
smallest cell n = 5 (NDB & severe)
a
n = 10
21.3%
[10.7, 35.7]
n = 17
36.2%
[22.7, 51.5]
n = 5
10.6%
[3.5, 23.1]
n = 15
31.9%
[19.1, 47.1]
DCB
NDB
efficacy (E)
severe irAE
no severe irAE
toxicity (T)
cell share
35%
30%
25%
20%
15%
Joint multinomial buys nothing out-of-sample
b
joint better in 12/25 folds
pooled log-loss
joint 1.3452 vs independent 1.3560
sign-permutation P = 0.986
Δ log-loss (joint − independent)
0.4
0.2
0.0
-0.2
1
5
10
15
20
25
CV fold (5-fold × 5 repeats, seed 20260911)
Identifiability of the joint estimand
no identifiable joint advantage at n = 47 — boundary, not discovery
c
events per parameter = 5.2 (47 / 9)
— far below the conventional 10
smallest cell n = 5 (NDB & severe)
smallest training cell n = 4 (every fold)
effective free parameters
15
10
5
0
6
9
independent pair
(2 logistics)
joint multinomial
(4 classes)
d
v82 boundary: E and T are measured in DIFFERENT cohorts
within-patient pairing unobservable → the joint (E,T) estimand
is not estimable from any current retrospective cohort
same patient
one baseline visit
(prospective)
tumour bulk RNA-seq → S0
PBMC scRNA → S0 (same draw)
paired (E, T) outcomes per patient
→ within-patient E–T coupling becomes estimable
prospective multi-cancer cohort, paired n powered for the joint estimand
(the frozen prediction system is never re-estimated; the bridge supplies only the pairing)